Psychrometric Tool

Application: HVAC · built Nov 12, 2026

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Heating, ventilation and air conditioning (HVAC) is the technology of indoor and vehicular environmental comfort. Its goal is to provide thermal comfort and acceptable indoor air quality. HVAC system design is a subdiscipline of mechanical engineering, based on the principles of thermodynamics, fluid mechanics, and heat transfer. Refrigeration is sometimes added to the field's abbreviation as HVAC&R or HVACR, or ventilating is dropped as in HACR (such as the designation of HACR-rated circuit breakers).

HVAC is an important part of residential structures such as single family homes, apartment buildings, hotels and senior living facilities, medium to large industrial and office buildings such as skyscrapers and hospitals, onboard vessels, and in marine environments, where safe and healthy building conditions are regulated with respect to temperature and humidity, using fresh air from outdoors.

HVAC engineers use psychrometrics to translate the knowledge of heating or cooling loads (which are in kW or tons) into volume flow rates (in m3/s or CFM) for the air to be circulated into the duct system. The volume flow rate is used to determine the size of fans, grills, outlets, air-handling units, and packaged units. This in turn affects the physical size (foot print) of air handling units and package units and is the single most important factor in conceptualizing the space requirements for mechanical rooms and also the air-distribution ducts. The main function of the psychrometric analysis of an air-conditioning system is to determine the volume flow rates of air to be pushed into the ducting system and the sizing of the major system components.